

FATIMA MUHAMMAD MUSHTAQ

📍 Islamabad, Pakistan

✉ fatima.mushtaq40@gmail.com



5/December/2003



[fatima@Linkedin.com](#)



+922 3305865867



[fatima.website](#)

EDUCATION

Bachelor of Science in Biotechnology

International Islamic University, Islamabad

Jan 2022 - 2026

Expected Graduation: Feb 2026 | **CGPA:** 3.95

- **Relevant Coursework:** Environmental Biotechnology, Microbial Biotechnology, Food Biotechnology, Biochemical Engineering, Microbiology, Molecular Biology, Virology, Industrial Biotechnology, Genomics & Proteomics, Ecology, Biodiversity & Evolution.

RESEARCH EXPERIENCE

Undergraduate Researcher - Final Year Project

International Islamic University Islamabad

September 2025- Ongoing

Supervisor: Dr. Bushra Uzair, Associate Professor

- Conducting an undergraduate research project on coliphage-based Biocontrol strategies against multidrug-resistant E. coli
- Isolating bacteriophages from wastewater-associated samples while developing skills in basic experimental design, laboratory documentation, and microbiological data interpretation

PROFESSIONAL EXPERIENCE

Laboratory Intern

National Institutes of Health (NIH), Islamabad

July 2025- Aug 2025

Nutrition Division (WHO Focal Point for Food Safety)

- Completed internship in food and water chemical and microbiological laboratories.
- Conducted sample handling and safety testing procedures while maintaining laboratory data and sample records.
- Performed microbiological testing of food and water samples, including pathogen and coliform detection, while gaining hands-on experience in sample preparation, microbial culture, and water analysis for food safety and water quality monitoring.

POSTER PRESENTATIONS

Research Bench to Market - Innovent 2025

Inter-university Biotech Competition

Theme: Circular Economy

- 2nd Place Winner | Led 4-member team | Primary presenter.
- Designed poster visuals and coordinated presentation Content

Interactive Research Display

International Islamic University Islamabad 2024

- Led team of 6 members, coordinating research, design, and presentation strategies.
- Synthesized and visualized complex research methodologies from published literature.
- Delivered presentation to academic panel as part of research exposure training.

ACADEMIC PROJECTS

Understanding Vitiligo: Exploring Self-Esteem & Quality of Life.

Immunology Course Project | IIUI | Spring 2024

Bacterial Isolation from Traditional Fermented Foods

Microbial Biotechnology Laboratory Project | IIUI | Fall 2024

SEMINARS & WORKSHOPS

Seminars:

- *Breast Cancer Awareness | Organizer | Oct 2024*
- *Biotechnology Career Campus: Finding your Way | Organizer | 2024*
- *Translating Bioinformatics Research into Clinical Practices | Organizer | 2024*

Workshops:

- *International E-Workshop on Molecular Informatics | IIUI 2024*
- *Hands-on microbiology workshop | IIUI 2024*
- *Bioethics and biosafety workshop | IIUI 2024*

ONLINE CERTIFICATIONS:

Coursera:

- *Bacteria and Chronic Infections | University of Copenhagen*
- *Antimicrobial Resistance: Theory and Methods | Technical University of Denmark (DTU)*
- *New Paradigms in Wastewater Management | Politecnico di Milano*
- *Algae Biotechnology | University of California San Diego*
- *Social Dimensions of Antimicrobial Resistance | University of Amsterdam*
- *Circular Economy: Sustainable Materials Management | Lund University*
- *Environmental Hazards and Global Public Health | University of Michigan*
- *Whole Genome Sequencing of Bacterial Genomes: Tools and Applications | Technical University of Denmark (DTU)*

Others:

- *Scientific Research Writing Course | Multiomics 2023*
- *Aspire Leaders Program | 2025*

AWARDS & ACHIEVEMENTS

- *2nd position | Inter-University Innovent 2025*
- *Merit-based laptop award | IIUI 2023*
- *Gold medalist | Highschool | 2022*

EXTRACURRICULAR ACTIVITIES

CLC Creative Leadership Conference 2.0

July 2024- Oct 2024

Ambassador & Internee

- *Served as conference ambassador while managing social media content for blogs, writing, and publications team.*

RWS charity organization

Feb 2024- July 2025

Volunteer

- Promoted organizational fundraising initiatives and raised donations for underprivileged individuals in the community

Scholarship Insights/ Global Scholarship Network

Sept 2023- April 2024

Volunteer

- Conducted outreach to international scholarship recipients and coordinated their participation as guest speakers and enhancing student exposure to global academic pathways

SKILLS

Technical & Research

- Laboratory techniques: Microbiological techniques.
- Basics of R & Python.

Software & Digital Tools

- Design: Canva.
- Office Suite: MS Word, Excel, PowerPoint.

Interpersonal Skills

- Team leadership.
- Project coordination and management.
- Public speaking and presentations.

Languages

- English
- Urdu (Native Speaker)

